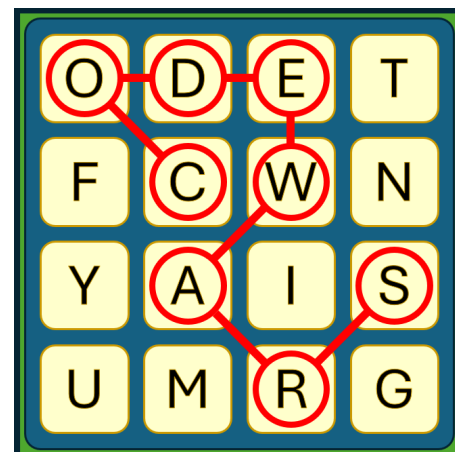


You're given a dictionary file words.txt. This is a dictionary from competitive Scrabble tournaments, there are nearly 200,000 words and lots of them are obscure. Your job is to read in all those words and then search a square letter grid from size 2x2 to size 10x10 for all the unique words of at least three letters from the dictionary that you can find in the grid. With 200,000 words and up to 100 starting points in the 10x10 grid, you'll need to be efficient with how you search. When forming a word you can use each letter in the grid only once and the letters must be vertically, horizontally, or diagonally adjacent. Print out the number of unique words followed by the list of words sorted alphabetically.



Input

The first line of input will be an integer which is the dimension of the letter grid. The following lines will be a grid of capital letters.

```
2
CO
DE
```

Output

Print out the number of unique words that are at least three letters long found in the dictionary file words.txt, followed by the list of words sorted in alphabetical order

```
8
COD
CODE
COED
DECO
DOC
DOE
ECO
ODE
```

Discussion

This is a more challenging problem than a straightforward word search. You'll need to be very efficient in how you generate all the possible words and how you check against the words in the dictionary. If you try to generate all the words and then check against the whole dictionary against each word, then your code won't be able to handle anything beyond the smallest grid sizes.

Be sure to run your solution against **all** of the student data sets.

Additional Examples

Input 2	Output 2
2 WA SR	8 ARS RAS RAW RAWS SAW WAR WARS WAS